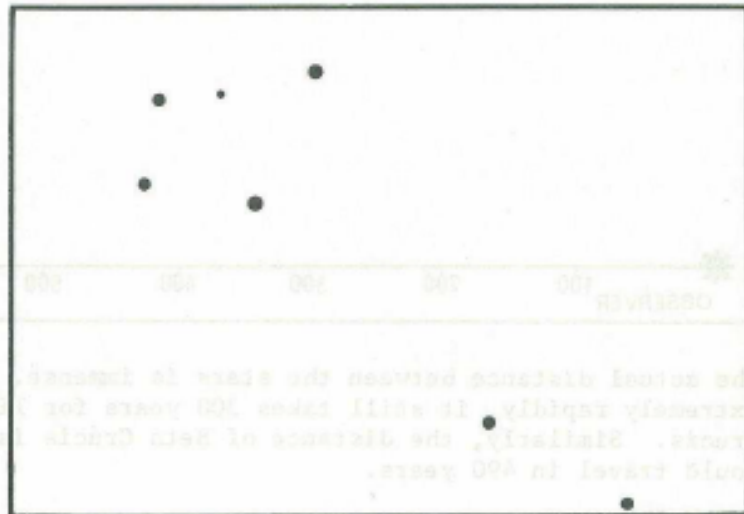


THE SOUTHERN CROSS

When we look at the Southern Cross in the sky it looks like the diagram on the right.

This is a *top view* of the constellation. It does not indicate the true positions of the stars in the sky relative to each other, for they all seem the same distance from the Earth. The stars of the Southern Cross are not closely associated with each other. They are in the same general direction but have very different distances.

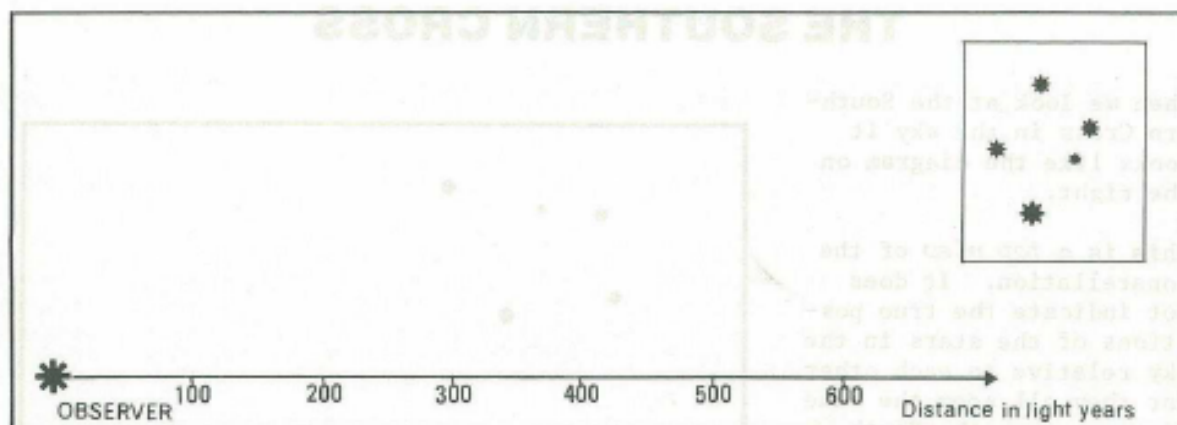


STAR	DISTANCE FROM EARTH
Alpha Crucis (α)	300 light-years
Beta Crucis (β)	490 light-years
Gamma Crucis (γ)	220 light-years
Delta Crucis (δ)	590 light-years
Epsilon Crucis (ε)	172 light-years

Over the page is a graph section which shows the Southern Cross on the right-hand side.

Draw a side view of the Southern Cross by completing the following

1. Draw a straight line from each star to the observer.
 2. Plot each star distance on the appropriate star line you have just drawn.
 3. Label each star with its Greek symbol.
- Explain the term, light-year

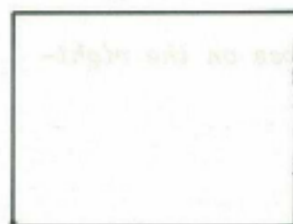


The actual distance between the stars is immense. Even though light travels extremely rapidly, it still takes 300 years for light to reach us from Alpha Crucis. Similarly, the distance of Beta Crucis is equal to the distance light would travel in 490 years.

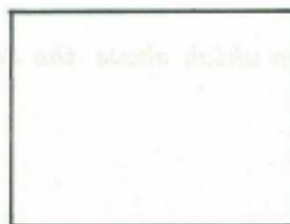
- If the stars of the Southern Cross were suddenly removed from space, we would not know for many years. Why?

- How long after the disappearance would we observe a change in the Southern Cross?

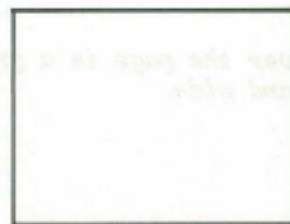
- Sketch what the Southern Cross would be like 250, 400 and 500 years after its disappearance.



250 years



400 years



500 years

The Sun is our own star. It is the closest star to Earth. Light, which travels at 300 000 kilometres per second, takes approximately 8 minutes to reach Earth.

- How far is Earth from the Sun?

The next closest star is Proxima Centauri, which is 4.2 light-years away.

- How far is this star from Earth in kilometres?